

Data, Signals and Events Used in Communication Failure

Type	Name	Meaning
Data	truckID	Identifies the affected truck
Data	linkStatus	Shows whether communication is available or failed
Data	failureType	Timeout, link loss, or message loss
Data	retryCount	Number of recovery attempts
Signal	heartbeat	Normal communication signal between trucks
Signal	failure_detected	Sent when communication timeout occurs
Signal	safe_mode	Activates safe state
Signal	recovery_request	Attempts to restore communication
Signal	recovery_success	Communication restored
Event	Timeout	No message received within allowed time
Event	Link Loss	Communication channel becomes unavailable
Event	Recovery Retry	Retry every 10 seconds
Event	Communication Restored	Truck returns to normal operation

Communication Protocol

Heartbeat-based V2V Communication Protocol.

For the communication failure scenario, a heartbeat-based V2V protocol is appropriate. Each truck periodically sends status messages. If no heartbeat is received within the timeout limit, the receiving truck detects communication failure, enters safe mode, and attempts recovery every 10 seconds.